

BIZPRO

Version 1.0 · Internal Document

BizPro is a custom ERP for a Delhi retail business with MULTIPLE shops and multiple warehouses. It covers inventory management, billing (with GST), orders, stock transfers, and user access control. This document explains the complete system flow, data collections, and field definitions.

>5-7 Shops	=>3+ Warehouses	38+ Vendors	>5 User Roles	4 Price Types
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1. System Overview & Architecture

BizPro is structured as a **centralized system with distributed access**. All data lives in one database, but each user sees only what their role allows. Stock updates happen in real-time — the moment a sale is made at any shop, inventory is updated everywhere.

Input Layer	Vendors, Purchase Orders, Bulk Upload	Getting stock into the system
Storage Layer	Warehouses, Rooms, Racks, Positions	Where physical stock lives
Movement Layer	WH→Shop, Shop→Shop, WH→WH transfers	How stock moves between locations
Sales Layer	Shops, Billing Counter, Orders	Where stock is sold to customers
Finance Layer	Bills, GST, Credit Notes, Payments	Money and records
Control Layer	User Roles, Access Control, Reports	Who can see and do what

2. Complete System Flow

The diagram below shows the full journey of a product — from when it arrives from a vendor to when it is sold to a customer and the inventory is updated.

1 Product Master Setup

- Every product is first registered in the system with its name, brand, category, HSN code, and 4 price types.
- 4 Price Types: MRP (walk-in customer), Wholesale (bulk buyer), Retail (regular shop), Online (e-commerce).
- GST % and GST type (CGST+SGST or IGST) are set per product for automatic billing.
- A system barcode is automatically generated for every new product — used for sticker printing and scanning.



2 Vendor Purchase & Goods Receipt

- When goods arrive from a vendor/ vendors, a purchase entry is created in the system.
- Each purchase entry records: vendor ID, invoice number, products, quantities, and purchase cost.
- The system generates a barcode sticker — printed and stuck on the physical product.
- If the manufacturer's barcode already exists on the product, it is mapped as an alternate barcode.



3 Warehouse Placement & Stock Location

- After receipt, every product is assigned a physical location: Warehouse → Room → Rack → Position.
- Scanning the barcode at any time shows exactly where the product is stored.
- Stock levels tracked: Total, Available, Reserved (for pending orders), In Transit (being moved).
- Low stock alerts fire automatically when quantity falls below the threshold set per product.



4 Distribution — Moving Stock to Shops

- Stock moves from Warehouse to Shop via a dispatch challan (pre-entry before goods physically leave).
- Shop-to-Shop transfers are also possible — only available quantity is visible, not sales data.
- Warehouse-to-Warehouse transfers happen with approval and full audit trail.
- Every movement (WH→Shop, Shop→Shop, WH→WH) is logged with timestamp, quantity, and who approved it.



5 Shop Operations & Billing

- Customer arrives at a shop. Billing staff scans product barcodes — items are added to the bill automatically.
- Staff selects bill type: GST Bill (with IGST/CGST), Non-GST Bill, or Credit Note redemption.
- GST % is auto-applied from product master. Staff selects which GST number to use (multiple supported).
- Bill is generated as a PDF with shop letterhead and sent to customer via WhatsApp (mobile number search).
- Payment is recorded. Shop stock is immediately deducted. Central inventory is updated in real-time.



6 Credit Notes & Returns

- Customer returns a product → Credit Note is generated (PDF sent to customer's WhatsApp).
- Stock is automatically added back to the shop's inventory.
- Credit note can be redeemed at any shop — looked up by customer's mobile number.
- Staff can attach a photo of the original physical note for verification.



After every sale, return, or transfer:

- Stock is updated at the specific shop or warehouse immediately.
- Central inventory reflects the change — all shops and warehouses see live numbers.
- Low stock alerts fire if quantity drops below the threshold.
- Super Admin can see the full picture across all locations at any time.

3. Data Collections & Fields

Below are all the data collections (database tables) the system needs, with their fields. **Required** fields must always have a value. **Optional** fields can be blank. **System** fields are auto-generated — users never fill these.

3.1 Products (Product Master)

One record per product type. Not per unit — per type (e.g., "Surf Excel 1 Kg" is one record).

product_id	System	Auto-generated unique ID. e.g. PRD-001
product_code	Required	Human-readable code assigned during bulk upload or manual entry
system_barcode	System	Auto-generated barcode for sticker printing and scanning
vendor_barcode	Optional	Manufacturer / vendor barcode if already printed on product (alternate ID)
name	Required	Product name in English or Hindi. e.g. Surf Excel 1 Kg
brand	Optional	Brand name. e.g. HUL, Amul, Samsung
description	Optional	Short product description. Max 150 characters
category	Required	Main category. e.g. FMCG, Grocery, Electronics
sub_category	Optional	Sub-category. e.g. Detergent, Dairy, Mobile Accessories
hsn_code	Required	6-digit HSN code — mandatory for GST billing and return filing
mrp	Required	Maximum Retail Price. The ceiling price
wholesale_price	Required	Price for bulk buyers / distributors
retail_price	Required	Standard shop price for walk-in customers
online_price	Optional	Price for e-commerce platforms. Blank if not sold online
purchase_cost	Optional	Cost from vendor. Internal only — never shown on customer bill
gst_percent	Required	GST rate as a number. e.g. 18. Free text — any future rate supported
gst_type	Required	CGST+SGST (same state) or IGST (inter-state) or Exempt
unit_of_measure	Required	How the product is counted. e.g. Pcs, Kg, Box, Ltr
weight_per_unit	Optional	Weight in kg per single unit — for shipping calculations
image_filenames	Optional	Comma-separated image filenames. e.g. PRD-001_1.jpg, PRD-001_2.jpg
is_active	System	true/false — soft delete. Inactive products hidden from billing

created_at	System	Timestamp when product was first added
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3.2 Vendors (Vendor Master)

One record per supplier. Vendor ID is used across purchase orders and inventory records.

vendor_id	System	Auto-generated unique ID. e.g. VND-001
company_name	Required	Supplier company or person name. e.g. HUL, Reckitt India
contact_person	Optional	Sales representative name
phone	Required	10-digit mobile number
whatsapp	Optional	WhatsApp number if different from phone
email	Optional	Email address for orders and invoices
gst_number	Optional	Vendor GST registration number — needed for purchase records
city	Required	City where vendor is located
address	Optional	Full postal address
is_active	System	true/false — whether vendor is currently supplying

3.3 Warehouses & Locations

Two connected collections: Warehouse (the building) and Stock_Location (room + rack inside it).

warehouse_id	System	Auto-generated. e.g. WH01, WH02
warehouse_name	Required	Name of the warehouse. e.g. Main Delhi Warehouse
address	Required	Physical address
city	Required	City
manager_name	Optional	Person responsible for this warehouse
room_zone	Required	Room or zone inside warehouse. e.g. A, Ground, Cold Room
rack_shelf	Required	Specific rack or shelf. e.g. R3, Shelf-2B
position	Optional	Further position within the rack if needed

3.4 Stock Ledger (Inventory Movement Log)

Every stock movement is recorded as a ledger entry — this is the heart of the inventory system. Current stock at any location is calculated from this log.

ledger_id	System	Auto-generated unique entry ID

product_id	Required	Which product moved
movement_type	Required	Type: PURCHASE / WH_TO_SHOP / SHOP_TO_SHOP / WH_TO_WH / SALE / RETURN
from_location	Required	Source: warehouse ID or shop ID (null for new purchases)
to_location	Required	Destination: warehouse ID or shop ID
quantity	Required	Number of units moved. Always positive
batch_number	Optional	Batch / lot number for traceability
expiry_date	Optional	Expiry date if applicable
vendor_id	Optional	Filled only for PURCHASE type movements
invoice_number	Optional	Vendor invoice number for purchase entries
approved_by	Optional	User ID who approved the transfer
created_by	Required	User ID who created this entry
created_at	System	Timestamp of the movement

3.5 Shops / Sales Offices

shop_id	System	Auto-generated. e.g. SHP-01
shop_name	Required	Display name. e.g. Karol Bagh Shop
address	Required	Physical address of the shop
city	Required	City
gst_numbers	Required	Array of GST numbers registered for this shop (multiple allowed)
bank_accounts	Required	Array of bank account details for bill dropdown
owner_user_id	Required	User ID of the shop owner
sales_channels	Optional	e.g. ["MHM", "OWB", "Walk-in"] — for platform-wise reports
is_active	System	true/false

3.6 Orders & Bills

One record per transaction. Each bill has line items (one per product) stored separately.

bill_id	System	Auto-generated bill number
shop_id	Required	Which shop generated this bill
bill_type	Required	GST_BILL / NON_GST_BILL / CREDIT_NOTE
customer_mobile	Optional	Customer mobile number — for WhatsApp delivery and credit note lookup

customer_name	Optional	Customer name
items	Required	Array of line items — see Bill Items below
subtotal	Required	Total before GST
gst_amount	Required	Total GST applied
total_amount	Required	Final amount customer pays
payment_method	Required	Cash / UPI / Card / Credit
gst_number_used	Optional	Which GST number of the shop was used for this bill
bank_account_used	Optional	Which bank account shown on bill
sales_channel	Optional	Platform: Walk-in / MHM / OWB / etc.
pdf_url	System	Link to generated PDF bill stored in system
created_by	Required	User ID of billing staff
created_at	System	Bill timestamp

Bill Items (one row per product in a bill):

bill_item_id	System	Auto-generated
bill_id	Required	Parent bill this item belongs to
product_id	Required	Which product was sold
quantity	Required	How many units sold
price_type	Required	Which price used: MRP / Wholesale / Retail / Online
unit_price	Required	Actual price per unit at time of sale
gst_percent	Required	GST % applied — copied from product master at sale time
line_total	Required	unit_price × quantity

3.7 Credit Notes

credit_note_id	System	Auto-generated
original_bill_id	Required	Bill this return is against
shop_id	Required	Shop where return happened
customer_mobile	Required	Mobile number — used to look up and redeem credit note
items_returned	Required	Array of products returned with quantities
credit_amount	Required	Total value of the credit note
status	System	ACTIVE / REDEEMED / PARTIALLY_REDEEMED
redeemed_at_shop	Optional	Shop ID where credit note was redeemed (if done)
photo_url	Optional	Photo of original physical credit note

created_at	System	Timestamp
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3.8 Users

user_id	System	Auto-generated
name	Required	Full name
phone	Required	Mobile number — used for login
role	Required	SUPER_ADMIN / WH_MANAGER / SHOP_OWNER / BILLING_STAFF / STOCK_LISTER
shop_id	Optional	Which shop this user belongs to (null for Super Admin and WH Manager)
warehouse_id	Optional	Which warehouse (for WH_MANAGER role)
is_active	System	true/false
created_at	System	Timestamp

4. Data Flow Diagrams (DFDs)

Each DFD below shows a specific process — what goes in, what happens, and what comes out. These are simplified Level-1 DFDs meant to be readable by non-developers.

DFD 1 — Goods Receipt from Vendor

Process: Vendor Purchase & Stock Entry		
Inputs	Processes	Outputs
• Vendor delivers goods	• Create purchase entry in system	• Stock added to warehouse
• Vendor hands over invoice	• Link to Vendor ID	• Purchase record saved
• Staff counts physical stock	• Assign warehouse location (Room + Rack)	• Barcode sticker printed
	• Generate barcode sticker	• Stock ledger entry created
	• Print and stick on product	
	• Update stock ledger	

DFD 2 — Stock Transfer (WH to Shop / Shop to Shop)

Process: Stock Movement Between Locations		
Inputs	Processes	Outputs
• Transfer request raised	• Check available stock at source	• Stock deducted at source
• Source location has stock	• Create dispatch challan	• Stock added at destination
• Receiving location confirmed	• Approve transfer (if WH→WH)	• Transfer log created
	• Scan barcodes at dispatch	• Both parties see updated qty
	• Receive and confirm at destination	
	• Update both location stock levels	

DFD 3 — Sale & Billing at Shop Counter

Process: Customer Sale & Bill Generation		
Inputs	Processes	Outputs
• Customer selects products	• Look up product from barcode	• Bill PDF created
• Staff scans barcodes	• Auto-apply price and GST	• Stock deducted at shop
• Payment method selected	• Calculate total and GST	• Central inventory updated
• Customer mobile (optional)	• Select bill type (GST / Non-GST)	• Payment recorded
	• Record payment	• Bill sent to customer
	• Generate PDF bill	

	• Send to customer WhatsApp	
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DFD 4 — Customer Return & Credit Note

Process: Return & Credit Note Lifecycle		
Inputs	Processes	Outputs
• Customer returns product	• Look up original sale	• Credit note created (ACTIVE)
• Original bill or mobile number	• Verify return eligibility	• Stock restored at shop
• Photo of physical receipt (optional)	• Create credit note	• PDF sent to customer
	• Add stock back to shop	• Credit available for future purchase
	• Send credit note PDF to WhatsApp	

DFD 5 — Real-Time Inventory Sync Loop

Process: Stock Update After Every Transaction		
Inputs	Processes	Outputs
• Any sale made at any shop	• Deduct / add qty in stock ledger	• Live stock count updated everywhere
• Any transfer completed	• Recalculate available stock at location	• Low stock alert (if triggered)
• Any return processed	• Check against low-stock threshold	• Super admin dashboard updated
	• Update central inventory view	• Shop-level stock reflects change
	• Fire alert if below threshold	

5. User Roles & Access Control

The system has 5 roles. Every screen and action in the system checks the user's role first. Users cannot access data outside their permission level.

Super Admin	✓ Full access to everything ✓ All warehouses + all shops ✓ All sales, inventory, billing data ✓ Full reports and analytics ✓ User creation and management ✓ System settings
Warehouse Manager	✓ Own warehouse only ✓ Stock in and out for their WH ✓ Transfer requests (send/receive) ✓ Cannot see shop sales data ✓ Cannot create bills
Shop Owner	✓ Own shop: full access ✓ Own sales + billing history ✓ Own stock levels ✓ Other shops: available qty only ✓ Cannot see other shops' sales
Billing Staff	✓ Own shop billing counter only ✓ Create GST / Non-GST bills ✓ Process returns / credit notes ✓ Cannot see inventory reports ✓ Cannot do transfers
Stock Lister	✓ Product listing only ✓ Bulk upload of inventory data ✓ No access to billing ✓ No access to sales data ✓ No transfer permissions

Key Privacy Rule: Shop-to-Shop stock visibility — a shop can see **how much stock** another shop has (to request a transfer), but **cannot see that shop's sales, pricing decisions, or transaction history**.

6. Barcode System

Every product in BizPro gets its own internal barcode. This barcode is printed on a sticker and attached to the product. All scanning — at warehouse receipt, during transfers, and at billing counters — uses this barcode.

Barcode Structure

Business Prefix	VYP	Fixed prefix for all BizPro barcodes — identifies the system
Warehouse Code	WH02	Where this product originally entered the system
Product Sequence	P00847	Auto-incrementing product type ID — same for all units of same product
Batch Number	003	Which purchase batch this unit came from — for traceability
Full Barcode	VYP-WH02-P00847-003	Complete barcode printed on sticker label

Barcode Format: Code 128

Code 128 is used — it supports letters and numbers, is compact, and scans fast on standard thermal label printers. If a manufacturer barcode already exists on the product, it is stored as an alternate ID. Both barcodes will work for scanning — the system recognises either.

When is a barcode generated?

Situation	What happens
Brand new product, first time ever	New product ID created (e.g. P00848). Batch starts at 001. New barcode generated.
Same product, new purchase batch	Product ID stays the same. Batch number increments (e.g. 001 → 002). New barcode for this batch.
Product has manufacturer barcode	Manufacturer barcode stored as alternate. System barcode also generated. Both work for scanning.

7. Key Business Rules

These rules must be enforced by the system at all times — they are not optional.

Inventory

- Stock can never go below zero. Any transaction that would cause negative stock must be blocked.
- If two billing counters try to sell the last unit simultaneously, only one succeeds — atomic transactions required.
- Every stock change (up or down) must create a ledger entry. No silent updates.
- Stock deduction at a shop happens immediately when payment is confirmed — not before.

Transfers

- A shop can see another shop's available quantity — but never its sales data or pricing decisions.
- Warehouse-to-Warehouse transfers require approval before stock changes hands.
- A dispatch challan must be created before goods physically leave a warehouse.
- In-transit stock is tracked separately — it is not available at source or destination until confirmed.

Billing

- GST % and HSN code are auto-pulled from the product master — billing staff cannot change them.
- A bill cannot be finalised without a payment method recorded.
- All bills are permanently stored — they can be viewed but not edited after creation.
- Credit notes can be redeemed at any shop in the network, not just where the return happened.

Pricing

- 4 price types exist per product: MRP, Wholesale, Retail, Online. MRP is always the ceiling.
- Wholesale price must always be less than or equal to MRP.
- Retail price must always be less than or equal to MRP.
- Billing staff selects which price type to use at time of sale — the system records which was used.

Access

- No user can access data outside their assigned role — enforced server-side, not just on the UI.
- Billing staff can only create bills — they cannot view inventory reports or do transfers.
- Super Admin is the only role that can see cross-location sales data.
- User accounts can be deactivated but not deleted — full audit trail is preserved.

End of Document — BizPro System Design v1.0